

Exploring Speech Emotion Recognition: From First Principles to State-of-the-Art Techniques

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Abstract— Speech Emotion Recognition is a vital field for understanding how humans perceive each other while communicating. Self-Supervised Learning (SSL) has introduced various models such as HuBERT and Wav2Vec 2.0 which show positive results in extracting meaningful representations from raw speech. Apart from these models, popular datasets such as IEMOCAP, CREMA-D, RAVDESS and SAVEE provide strong benchmarks for evaluation. This survey aims to analyze the architecture, methodology and fine-tuning methods of HuBERT and wav2vec 2.0 to understand their effectiveness in capturing linguistic and para-linguistic features from speech. The datasets are analyzed for their attributes such as gender and ethnic diversity, methods of annotation, amount of speech hours and the nature of their recordings which contribute significantly to their usability. The comparative results highlight the strengths, limitations, and performance of each model which offers insights into the trade-offs between the architecture of the models and the various properties of the datasets. These results provide a strong base for advancing research in SER.

Keywords—Speech Emotion Recognition, Self-Supervised Learning, HuBERT, Wav2Vec 2.0

I. INTRODUCTION

Speech carries useful information beyond lexical content which reflects attitude, interpersonal dynamics and a wide range of other emotional states commonly ignored by popular voice assistant models such as Amazon Alexa, Google Assistant and Apple Siri. These voice assistants focus on Automatic Speech Recognition (ASR) which works by transcribing spoken words into text and ignoring the vocal cues hidden in tone, pitch, rhythm and intensity. These cues are often more useful than the words themselves as they provide valuable insights into whether a speaker is confident, hesitant, frustrated or engaged.

The ability to capture such meaningful insights forms the foundation of Speech Emotion Recognition (SER). In this survey, we provide a summary of recent developments in SER, emphasizing state-of-the-art self-supervised acoustic models such as HuBERT (Hidden Units Bidirectional Encoder Representations from Transformers) and Wav2Vec 2.0, as well as some fine-tuned models designed specifically for emotion recognition tasks. In addition to these models, we also examine the role of popular datasets in shaping SER progress. Datasets such as IEMOCAP, CREMA-D, and RAVDESS differ in emotional granularity, annotation schemes, and recording conditions, all of which directly impact model training and evaluation.

Self-Supervised Learning (SSL) has proven to be an effective method for learning acoustic representations [1], [2] while overcoming common issues that come with datasets such as annotation costs, annotation bias, lack of domain generalization, and the lack of robustness. Unlike traditional supervised approaches, which require extensive labelled data, SSL leverages large volumes of raw, unlabelled speech to learn meaningful features. By designing auxiliary or pretext tasks, such as reconstructing masked portions of an audio signal [3], predicting future frames, or distinguishing between true and corrupted speech segments [4], SSL enables models to capture phonetic, prosodic, and semantic patterns without relying on human annotations.

II. SER IN HUMANS

Humans identify emotions in speech through acoustic, visual, and prosodic cues. Acoustic features such as pitch, intensity and duration provide important signs about emotional states [5]. Visual information like facial expressions, gestures and body language further enhance emotion recognition by complementing the auditory channel. Prosodic cues such as rhythm, stress patterns and intonation add an additional layer of meaning. Other physiological and invisible factors such as heart rate, blood pressure, skin conductance and breathing patterns add further to the multimodality of data.

A. Physiology of Speech Recognition

The human brain is central in decoding both the linguistic and emotional aspects of speech. The left hemisphere is responsible for processing grammar, semantics and syntax of speech while the right hemisphere processes paralinguistic features such as tone, pitch variation and emotional nuance.

Speech perception begins with the auditory pathway, where sound waves are captured by the outer ear and transmitted via the middle ear to the cochlea in the inner ear. The cochlea converts these vibrations into neural signals through the action of hair cells, which then transmit information via the auditory nerve to the brainstem. From there, signals are relayed to the primary auditory cortex (PAC) located in the temporal lobe, where basic sound features such as frequency and intensity are analysed. The amygdala and hippocampus allow for the emotional colouring of speech, linking auditory input with emotional memory and affective processing. This complex neural pathway and the multimodality of human speech emotion recognition form the foundation that current SOTA models try to replicate. These

models try to mimic the brain’s ability to integrate multiple signals, such as acoustic features, linguistic content, and contextual cues into a unified representation.

B. Relating AI Models and Humans

In models such as HuBERT and Wav2Vec 2.0, the lower layers (e.g., convolutional layers in Wav2Vec 2.0 or the transformer encoder base in HuBERT) extract low-level acoustic features (MFCCs, spectrogram-like patterns), similar to early cortical filtering in humans. In the brain, , located in the Superior Temporal Gyrus (STG) of the temporal lobe, processes pitch, amplitude (loudness), timbre, and spectral content which forms the foundation of auditory perception [5], [6].

The middle layers of these models (Wav2Vec 2.0 transformer layers 5–12 or HuBERT clustered units) capture phonetic sequences and prosodic patterns through contrastive or masked prediction [7]. Similarly auditory association areas like the posterior STG and planum temporale, integrate acoustic information with limbic structures such as the amygdala (emotional valence) and insula (arousal) [8]. These regions process prosody (intonation, rhythm), voice quality (breathiness, harshness), and paralinguistic cues [9].

At the upper layers, SER models form abstract and context-aware representations that resemble semantic representations, which can support multitask applications such as multimodal SER when combined with text [10]. In humans, this corresponds to the Prefrontal Cortex (PFC), Inferior Frontal Gyrus (IFG/Broca’s area, and Orbitofrontal Cortex, where auditory input is integrated with semantics, memory, and social context [11] using the help of attention networks.

Both HuBERT and Wav2Vec 2.0 rely on SSL, contrastive learning in Wav2Vec 2.0 and masked unit prediction in HuBERT, to learn hierarchical speech representations from vast amounts of unlabelled audio [4], [12]. This hierarchical structure parallels cortical predictive coding in the human brain, where incoming auditory signals are continuously compared to prior expectations and updated via error signals [11].

III. AUDIO FEATURES FOR SER

Speech Emotion Recognition (SER) is the process of automatically detecting and interpreting emotions from speech signals [12]. Audio features play a crucial role in SER as they capture the different elements through which emotional nuances are expressed in human speech. Human emotions can be represented in two ways:

1) *Categorical Emotions*: This represents discrete, distinct categories of emotions which are dependent of different audio features. The distinct category is decided by the pitch, intensity, speaking rate and other audio features. Ex: Happy, Sad, Anger, Fear, etc.

2) *Dimensional Emotions*: This represents emotions along continuous scales or dimensions. This approach is guided by three primitives, namely Arousal which refers to intensity of emotion (calm or excited), Valence which describes the pleasantness or unpleasantness, and Dominance, which indicates the level of control or submissiveness. Emotions are described by their position in multi-dimensional

space along the axes such as Arousal, Valence and Dominance(AVD). Emotions are categorized according to the AVD scale.

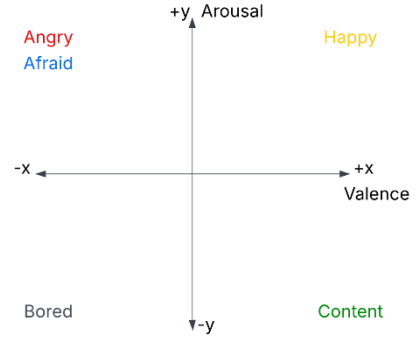


Figure 1. Representing emotions based on Valence and Arousal

A. Types of Audio Features

It focuses on the audio features on which the model can learn and generalize the type of emotion it is associated with. Acoustic captures the vocal qualities like frequency and pitch, Linguistic analyse the sentence structure and word choice to gain semantic insights, Prosodic captures delivery patterns like rhythm and intensity.

IV. CLASSICAL APPROACHES

Earlier Speech Emotion Recognition (SER) systems included both traditional both Machine Learning and initial Deep Learning techniques. Classical models such as K-Nearest Neighbors (KNN), Gaussian Mixture Models (GMMs), Support Vector Machines (SVMs) and Hidden Markov Models (HMMs) all are dependent on handcrafted acoustic features like MFCCs, LPC, pitch, intensity, and speaking rate. While the GMMs and HMMs effectively captured temporal features, SVMs performs better than simpler models such as KNN in particular for detecting sadness [13]. However, these methods struggle to generalize across different speakers and noisy conditions. With more computational power and data availability, deep architectures like CNNs, RNNs, LSTMs, DBNs, and Autoencoders advances in the field by learning hierarchical representations directly from raw audio or spectrograms. CNNs achieved over 90% accuracy on EmoDB and IEMOCAP, while RNNs and LSTMs captured temporal features with hybrid CNN-LSTM models surpassing the 80% accuracy, highlighting the advantages of combining spatial and sequential modeling [12], [13].

V. SELF-SUPERVISED LEARNING

Self-Supervised Learning (SSL) has emerged as a promising approach to address the challenges associated with traditional machine learning methods, particularly the dependence on large amounts of labeled data [7]. In many real-world applications, labeling data is expensive, time-consuming, and often impractical, especially when dealing with unstructured data like text, images, audio, or video. SSL reduces this dependency by leveraging unlabelled data to learn useful representations, making it a key step toward scalable and cost-efficient AI systems.

SSL training typically consists of two main phases: pretraining and downstream fine-tuning. These stages work together to build powerful and adaptable machine learning models.

Pretraining Phase: The pretraining phase is the core of self-supervised learning. In this process, the model picks up general patterns from a vast amount of unlabelled data by tackling a task that's automatically created, often referred to as a pretext task. The system generates pseudo-labels straight from the raw data. For instance, in BERT, the model randomly masks certain words in a sentence and trains itself to predict those missing words. This approach allows it to capture the patterns, structures, and subtleties within the data without needing any human input. While this phase can be quite demanding in terms of computation, it's scalable since it doesn't rely on labelled data, making it super-efficient for managing huge datasets like extensive text or audio collections.[5], [7].

Once the model has gone through the pretraining phase, we move on to the downstream phase, where it gets tailored for specific real-world tasks. This step involves using a smaller labelled dataset to fine-tune the model for a particular application. The general knowledge gained during pretraining is preserved, and a task-specific layer is added, allowing the model to be fine-tuned with the labelled data. The aim here is to refine the broad knowledge acquired during pretraining so that it performs exceptionally well on a specific task, even with minimal labelled data. [7].

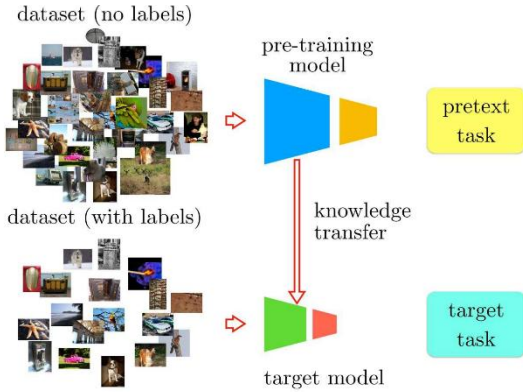


Figure 2. Downstream Task in SSL

A. Contrastive Learning

Contrastive learning is a self-supervised method learns meaningful representations by bringing similar instances (positive pairs) closer together in the embedding space while pushing apart dissimilar ones (negative pairs). The process usually includes data augmentation to create varied views, an encoder network (like CNNs for images or RNNs for sequences) to extract high-level features, and a projection network to fine-tune the embeddings. A contrastive loss function, which can be measured through cosine similarity or Euclidean distance, directs the training by maximizing the similarity of positive pairs and minimizing that of negative pairs. Models are optimized using algorithms like SGD on large-scale unlabelled datasets. The representations learned can be effectively transferred to related tasks, with performance on unseen data serving as a good indicator of how well the model generalizes.

VI. COMPARING MODELS

Here we compare two State of the Art pretrained models for generating embeddings for audio input, considering waveform as the time-series data.

A. HuBERT

A HuBERT (Hidden-Unit BERT) is a self-supervised model designed for learning speech representations. It's all about pulling out both acoustic and linguistic features straight from continuous speech. What sets HuBERT apart from traditional models is that it doesn't need explicit phoneme segmentation or a pre-set sound unit lexicon. Instead, it predicts masked areas of hidden units, much like how BERT handles masked language modeling. One of HuBERT's standout features is its focus on maintaining consistency across clustering iterations, which makes it really effective for a range of speech-related tasks, including Speech Emotion Recognition (SER), automatic speech recognition, and speaker verification. [5], [14].

The HuBERT-CLAP framework introduces a contrastive language-audio self-alignment pre-training approach specifically designed for SER [5]. This framework consists of two main stages.

In the first stage, known as contrastive pre-training, HuBERT processes audio inputs to create speech embeddings, while BERT or its lighter cousin, DistilBERT, works on text transcripts to pull out textual embeddings. These embeddings are then aligned using the CLIP model, which calculates scaled pairwise cosine similarities and applies a symmetric cross-entropy loss to ensure that the audio features closely match their corresponding textual features. This results in a shared multimodal representation space that effectively captures emotional cues from both audio and text. [5].

In the second stage, which zeroes in on downstream SER tasks, only the HuBERT part of the framework is partially fine-tuned. This setup makes the final system unimodal during inference, meaning it only needs audio input while still reaping the benefits of the multimodal pre-training phase. The fine-tuning process involves updating specific components like the feature projection layer, Transformer blocks, and the classifier head, while keeping the text model and CLIP frozen. This approach significantly cuts down on the computational

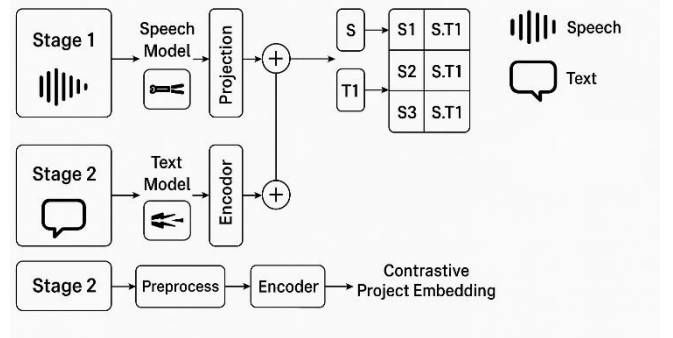


Figure 3. Architecture of HuBERT CLAP
costs associated with training and inference.[5].

Several HuBERT configurations were compared in ablation study on the IEMOCAP dataset for emotion recognition. The weighted accuracy (WA) of 70.04 percent was achieved by only using HuBERT without pre-training. The performance increased to 77.22 percent WA when

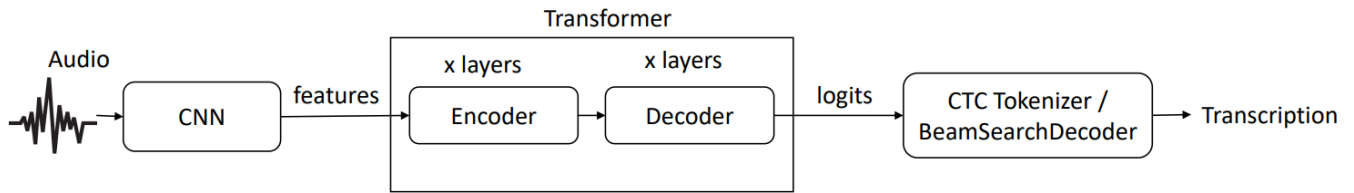


Figure 4. Architecture of Wav2vec 2.0

integrated with BERT pre-training and 76.91 percent WA when integrated with BERT’s lighter model DistilBERT pre-training [5]. On the same dataset, CNN+Bi-GRU (2019) achieved 70.39 percent WA, SPU+MSCNN (2021) scored 66.60 percent, LightSER (2022) scored 72.63 percent, and TIM-Net (2023) scored 71.65 percent when compared to recent state-of-the-art techniques. In contrast, HuBERT-CLAP (2024) outperformed all these models, reaching 77.22% WA, establishing a new benchmark on the IEMOCAP dataset [5].

1) Unimodal Approach

The unimodal approach utilizes a single input modality, which can be either audio or text, for Speech Emotion Recognition (SER). For the audio modality, HuBERT embeddings or baseline low-level acoustic features such as MFBF0 (mel-filterbank energies combined with pitch features) are employed to capture speech characteristics, including vocal tone, pitch, and energy variations. These features focus primarily on the acoustic properties of speech to infer emotional states [14].

For the text modality, BERT embeddings are generated from ASR-transcribed speech, allowing the system to capture rich semantic and contextual language cues present in the spoken content [15].

2) Multimodal Approach

The multimodal approach integrates both audio and text modalities, leveraging their complementary strengths to achieve more accurate and robust emotion recognition [14]. HuBERT acoustic embeddings are used to capture vocal tone, energy, and acoustic nuances, providing insights into how emotions are expressed through speech. BERT lexical embeddings capture semantic and contextual information derived from speech transcripts, offering a deeper understanding of the linguistic content [5], [14].

The two modalities are projected into a shared 128-dimensional space, after which they are concatenated and processed through the network. This fusion method significantly enhances SER performance. Notably, a 100% relative improvement in concordance correlation coefficient (CCC) for valence estimation was achieved compared to unimodal acoustic feature models.

B. Wav2Vec 2.0

Wav2Vec 2.0 is a self-supervised learning (SSL) framework developed for speech representation learning. Unlike traditional supervised automatic speech recognition (ASR) systems that require large amounts of labelled data, Wav2Vec 2.0 leverages unlabelled raw audio to learn meaningful speech representations [4]. It is then fine-tuned with a smaller labelled dataset for downstream tasks such as ASR, speech translation, and speech emotion recognition (SER). Additionally, the model can run on resource-

constrained devices such as Raspberry Pi, offering benefits such as privacy, reduced latency, and independence from internet connectivity [6], [16].

The feature encoder is a multi-layer convolutional neural network (CNN) that processes raw 16 kHz audio waveforms into low-level latent speech representations. It consists of seven temporal convolution blocks with the following configurations: Strides: (5, 2, 2, 2, 2, 2, 2), Kernel widths: (10, 3, 3, 3, 3, 2, 2). Each block contains 512 channels, enabling the extraction of robust representations from continuous speech [16].

The sequence of latent features is masked before being fed into a 12-layer transformer network in the base model. The transformer parameters are as follows: Model dimension: 768, Feed-forward dimension: 3072, Attention heads: 8. This module builds contextualized representations by attending to surrounding time steps, similar to the masking mechanism used in BERT for natural language processing [6].

The quantization module discretizes the latent speech representations into a trainable codebook. During pretraining, the model predicts masked quantized units using a contrastive loss, which distinguishes true masked targets from distractor samples [6].

During pretraining, a subset of the latent speech representations is masked, and the model learns to predict the masked segments using the surrounding context. This process enables Wav2Vec 2.0 to capture both phonetic and semantic patterns without requiring transcriptions. As a result, it is highly effective in low-resource settings, where labeled data is limited [6].

Finetuning Methods or approaches for wav2vec2.0 are Vanilla Fine-Tuning, Task-Adaptive Pretraining and Pseudo-Label Task-Adaptive Pretraining

VFT is the simplest and most direct fine-tuning approach for Wav2Vec 2.0. In this method, Wav2Vec 2.0 generates frame-level contextualized representations from raw audio inputs [4]. However, for utterance-level tasks such as Speech Emotion Recognition (SER), these frame-wise features must be aggregated into a fixed-length representation [4], [6]. To achieve this, Global Average Pooling (GAP) is applied across the time dimension to summarize the features into a single vector [4].

A significant challenge in adapting Wav2Vec 2.0 to SER is the issue of domain shift, which arises due to the mismatch between the pretraining data and the target data. The model is originally pretrained on neutral speech from ASR corpora such as LibriSpeech, whereas the target data consists of emotive speech characterized by variations in prosody, pitch, and intensity [4].

To tackle this discrepancy, Task-Adaptive Pretraining (TAPT) introduces an intermediate training phase. During this

phase, the Wav2Vec 2.0 model undergoes extra self-supervised pretraining on the target dataset (for example, IEMOCAP or SAVEE) without relying on labels, sticking to the original pretraining goal. Once this domain adaptation step is wrapped up, the model is fine-tuned using the standard procedure outlined in VFT [4].

The Pseudo-Label Task-Adaptive Pretraining (P-TAPT) method builds on TAPT by adding pseudo-labels to steer the model during pretraining. This technique is crafted to tap into frame-level emotional patterns, even when we only have utterance-level emotion labels at our disposal. The process kicks off with frame-level emotion extraction, where the original Wav2Vec 2.0 model is fine-tuned to predict frame-level emotion states using CNN-based architectures, ensuring the temporal structure of speech is maintained [4].

Next, k-means clustering is applied to the extracted frame-level representations to group similar emotional patterns. The resulting cluster assignments serve as pseudo-labels that represent local emotion states within the utterance [4]. The pretraining objective is then adjusted; instead of reconstructing masked audio segments like in TAPT, P-TAPT trains the model to predict the cluster IDs of masked frames [4]. To boost robustness, multiple cluster configurations are employed simultaneously to create a cluster ensemble, a strategy inspired by the HuBERT model [4].

Finally, after finishing the pseudo-label-based pretraining stage, a standard fine-tuning step is carried out using the actual utterance-level emotion labels. This approach allows the model to effectively capture both local frame-level and global utterance-level emotions.[4].

When it comes to IEMOCAP and SAVEE, the self-supervised representations from Wav2vec really shine, outperforming the traditional spectrogram-based methods. Interestingly, Wav2vec 2.0 manages to deliver similar performance whether it's fine-tuned or not. Plus, using parameter-efficient strategies takes things up a notch, with P-TAPT hitting an impressive 74.3% UA, outdoing both TAPT at 73.5% and the standard fine-tuning at 69.9%. On top of that, blending different modalities boosts recognition rates significantly. For instance, the combination of Audio and Text (MFCC+ALBERT) achieves a UA of 721%, while Audio and ASR (MFCC+BERT) reach 75.9%. This clearly shows that merging acoustic and linguistic elements leads to better results than sticking to just one approach.

VII. OVERVIEW OF SPEECH EMOTION DATASETS

The success of speech emotion recognition (SER) models largely depends on the quality of the speech databases (corpus) used, which are evaluated based on factors such as diversity of speakers, number of actors involved, amount of data available, and labelling methodology. Since emotion labeling is subjective, it is often performed by multiple annotators, either trained experts or untrained listeners, to ensure reliability and reduce individual bias. The nature of the dataset also plays a crucial role: synthetic datasets (e.g., RAVDESS, Emo-DB) rely on acted emotions in controlled environments, semi-natural datasets combine acted and spontaneous elements (e.g., NIMITEK), and natural datasets capture real-life emotional speech from sources such as call centers, TV shows, or online media.

A. IEMOCAP

The Interactive Emotional Dyadic Motion Capture (IEMOCAP) database is a multimodal corpus developed by the Speech Analysis and Interpretation Laboratory (SAIL) at the University of Southern California (USC) [13]. The corpus was recorded by 10 actors, 5 female and 5 male. It contains approximately 12 hours of audiovisual recordings, including speech, text transcripts, video, and motion capture data of facial expressions and hand movements. The dataset consists of dyadic (two-person) sessions, where professional actors perform both scripted and improvised scenarios designed to elicit a range of emotions. These include categorical emotions such as anger, happiness (often merged with excitement), sadness, neutral, and others, as well as dimensional annotations based on the Valence, Activation, Dominance (VAD) model [17], [18].

Due to its size, quality, multimodal design and comprehensive annotations, IEMOCAP has become one of the most widely used benchmarks in Speech Emotion Recognition (SER) research. It has been employed in a large proportion of studies, appearing in 30 out of 51 total referenced works, including 30 out of 39 English-language SER studies [12].

B. CREMA-D

Crowd-Sourced Emotional Multimodal Dataset (CREMA-D) is a large-scale audiovisual corpus consisting of 7,442 clips performed by 91 actors (48 male and 43 female) from diverse ethnic and cultural backgrounds across Asia, Africa, Europe, and the Americas [19]. The dataset was manually evaluated by 2443 participants achieving different results for audio-only, visual-only, and audio-visual data which are 40.9, 58.2 and 63.6 percent respectively. In addition to this, the recognition rates also differ for every emotion where neutral achieved the highest followed by happiness, anger, disgust, fear, and sadness [20].

C. RAVDESS

The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) is a gender-balanced dataset consisting of 24 professional North American actors [21]. The speech component includes lexically matched sentences, meaning that all sentences use the same set of words, ensuring consistency in vocabulary rather than just similarity in meaning. Sentences were recorded at two levels of emotional intensity in addition to a neutral expression. The database consists of 7356 recordings each rated 10 times for emotional validity and genuineness. The ratings were initially given by 246 untrained individuals and then further rated by another 72 individuals to confirm reliability. This dataset is publicly available making it a suitable choice for SER tasks [22].

D. SAVEE

Surrey Audio-Visual Expressed Emotion (SAVEE) database is an audio corpus recorded at the University of Surrey in 2010 [23], [24]. It consists of only 4 native English speakers each with their own label (DC, JE, JK and KL). The database captures seven emotions: happy, pleasant surprise, sad, neutral, anger, disgust and fear. However due to its small sample size, restricted access and limited demographic diversity, SAVEE has seen less use in speech emotion recognition (SER) research compared to larger, more diverse datasets like IEMOCAP and RAVDESS.

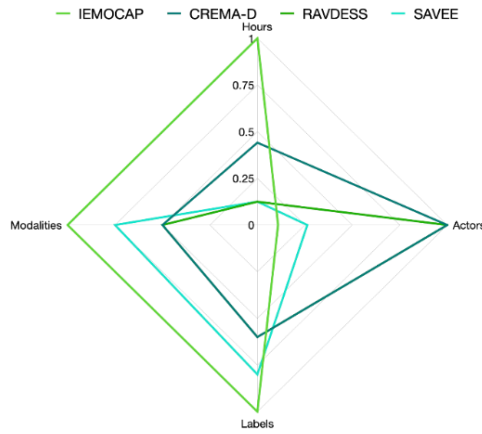


Figure 5. Summary of parameters of all datasets

VIII. CONCLUSION

In this study, we have understood human anatomy, techniques, models and datasets which are commonly involved with SER. The auditory pathway and the various parts of the human brain involved with processing speech and understanding emotions have also been presented. HuBERT is a multimodal model which leverages both text and audio to understand speech whereas Wav2Vec 2.0 uses only audio which makes each model suitable for different applications where it has been seen that fine tuning them gives significantly improved results. When it comes to models, IEMOCAP stands out as the most popular despite its restricted access due its ethnic and gender diversity, amount of recording hours and the nature of each recording which consists of both scripted and improvised dialogue which is crucial for models to extract emotional cues from. CREMA-D provides the highest amount of data but suffers from crowd-sourced labelling which affects its reliability. RAVDESS is publicly available and frequently used but lacks diversity in actors. SAVEE is constrained by its lack of both gender and ethnic diversity and restricted access. These comparisons emphasize the trade-offs in models and datasets commonly found in SER and relating the anatomy of humans with models provide a helpful guide for researchers in choosing the right resources for each application.

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